

Lecture Scribe Tutorial

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Use strictly and exclusively the provided context, which consists of lecture slides/PDF and relevant textbook excerpts. Every statement, equation, definition, derivation, and example must be traceable to the provided context. Do not introduce any external knowledge, additional examples, interpretations, or simplifications.

The lecture scribe must faithfully reconstruct the lecture content and include the following components whenever they appear in the provided context:

- Definitions and notation exactly as presented
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- Statements of theorems, results, or properties
- Proofs or derivations reproduced in complete step-by-step form
- Worked examples including all intermediate steps
- Logical progression and dependencies between concepts
- Mathematical expressions written clearly and accurately

Maintain strict correctness and completeness. Do not omit intermediate steps in proofs, derivations, or example solutions if they appear in the lecture materials.

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Do NOT include:

- New examples, exercises, or applications
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Ensure the final output is precise, structured, complete, and fully aligned with the provided lecture and textbook content.



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All definitions, derivations, and results are reproduced faithfully with complete steps and no external additions.

CSE 400: Fundamentals of Probability in Computing

Tutorial 2 — Lecture Scribe (Authoritative Exam Reference)

Q1. Random Point on a Line Segment

Problem Statement

A point is chosen at random on a line segment of length L . Find the probability that the ratio of the shorter to the longer segment is less than $\frac{1}{4}$.

Mathematical Formulation

Let the point be at distance x , where:

$$0 < x < L$$

Segments formed:

$$x \quad \text{and} \quad L - x$$

Ratio:

$$R = \frac{\min(x, L - x)}{\max(x, L - x)}$$

Case Analysis

Case 1: $x \leq \frac{L}{2}$

$$\frac{x}{L-x} < \frac{1}{4} \Rightarrow 4x < L-x \Rightarrow 5x < L \Rightarrow x < \frac{L}{5}$$

Case 2: $x > \frac{L}{2}$

$$\frac{L-x}{x} < \frac{1}{4} \Rightarrow 4L-4x < x \Rightarrow 5x > 4L \Rightarrow x > \frac{4L}{5}$$

Valid Region

$$(0, \frac{L}{5}) \cup (\frac{4L}{5}, L)$$

Total length:

$$\frac{L}{5} + \frac{L}{5} = \frac{2L}{5}$$

Final Probability

$$P = \frac{2L/5}{L} = \frac{2}{5}$$

Q2. Bayesian Classification with Normal Distributions

Given

- $X|W \sim N(4, 4) \Rightarrow \sigma_1 = 2$
 - $X|B \sim N(6, 9) \Rightarrow \sigma_2 = 3$
 - $P(B) = \alpha, \quad P(W) = 1 - \alpha$
 - Observation: $X = 5$
-

Condition for Equal Error

$$P(B|X = 5) = P(W|X = 5) = \frac{1}{2}$$

Bayes' Theorem

$$\frac{\alpha f(5|B)}{\alpha f(5|B) + (1 - \alpha) f(5|W)} = \frac{1}{2}$$

Normal Densities

$$f(x) = \frac{1}{\sigma \sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$
$$f(5|B) = \frac{1}{3 \sqrt{2\pi}} e^{-1/18}, \quad f(5|W) = \frac{1}{2 \sqrt{2\pi}} e^{-1/8}$$

Substitution

$$\frac{\alpha e^{-1/18} / (3 \sqrt{2\pi})}{\alpha e^{-1/18} / (3 \sqrt{2\pi}) + (1 - \alpha) e^{-1/8} / (2 \sqrt{2\pi})} = \frac{1}{2}$$

Simplified Equation

$$\frac{3\alpha}{3\alpha + 2(1 - \alpha)e^{-5/72}} = \frac{1}{2}$$

Solving

$$6\alpha = 3\alpha + 2(1 - \alpha)(0.93)$$

$$3\alpha = 1.86 - 1.86\alpha$$

$$4.86\alpha = 1.86 \Rightarrow \alpha \approx 0.3827$$

Final Result

$$\alpha \approx 0.38$$

Q3. Optimal Location Problem

(a) Uniform Distribution

Given

$$X \sim \text{Uniform}(0, A), \quad f_X(x) = \frac{1}{A}$$

Objective

Minimize:

$$E[|X - a|]$$

Expression

$$E[|X - a|] = \int_0^a (a - x) \frac{1}{A} dx + \int_a^A (x - a) \frac{1}{A} dx$$

Derivative

$$\frac{d}{da} \left(\frac{2a^2 - 2aA + A^2}{2A} \right) = \frac{1}{2A} (4a - 2A)$$

Optimal Point

$$4a = 2A \Rightarrow a = \frac{A}{2}$$

Result

Station at midpoint.

(b) Exponential Distribution

Given

$$X \sim \text{Exponential}(\lambda), \quad f_X(x) = \lambda e^{-\lambda x}$$

Derivative

$$\frac{d}{da} E[|X - a|] = 1 - 2e^{-\lambda a} = 0$$

Solving

$$e^{-\lambda a} = \frac{1}{2} \Rightarrow -\lambda a = \ln(1/2) \Rightarrow a = \frac{\ln 2}{\lambda}$$

Q4. Mixture of Exponential Lifetimes

Given

- Type i : rate λ_i
 - Probability p_i
-

Total Probability

$$P(X > t) = p_1 e^{-\lambda_1 t} + p_2 e^{-\lambda_2 t}$$

$$P(X > t + s) = p_1 e^{-\lambda_1(t+s)} + p_2 e^{-\lambda_2(t+s)}$$

Conditional Probability

$$P(X > t + s | X > t) = \frac{p_1 e^{-\lambda_1(t+s)} + p_2 e^{-\lambda_2(t+s)}}{p_1 e^{-\lambda_1 t} + p_2 e^{-\lambda_2 t}}$$

Q5. Conditional Distribution of Poisson Variables

Given

$$X \sim \text{Poisson}(\lambda_1), \quad Y \sim \text{Poisson}(\lambda_2)$$

Conditional Probability

$$P(X = k | X + Y = n) = \frac{P(X = k)P(Y = n - k)}{P(X + Y = n)}$$

Substitution

$$= \frac{e^{-\lambda_1} \lambda_1^k / k! \cdot e^{-\lambda_2} \lambda_2^{n-k} / (n-k)!}{e^{-(\lambda_1 + \lambda_2)} (\lambda_1 + \lambda_2)^n / n!}$$

Final Form

$$P(X = k | X + Y = n) = \binom{n}{k} \left(\frac{\lambda_1}{\lambda_1 + \lambda_2} \right)^k \left(\frac{\lambda_2}{\lambda_1 + \lambda_2} \right)^{n-k}$$

Q6. Transformation of Random Variables

Given

$$X \sim \text{Uniform}[-2, 2], \quad f_X(x) = \frac{1}{4}$$

$$g(x) = \begin{cases} x, & [-2, -1] \\ 0, & (-1, 1) \\ x, & [1, 2] \end{cases}$$

CDF $F_Y(y)$

$$F_Y(y) = \begin{cases} 0, & y < -2 \\ \frac{y+2}{4}, & -2 \leq y < -1 \\ \frac{1}{4}, & -1 \leq y < 0 \\ \frac{3}{4}, & 0 \leq y < 1 \\ \frac{y+2}{4}, & 1 \leq y \leq 2 \\ 1, & y > 2 \end{cases}$$

Point mass:

$$P(Y = 0) = \frac{1}{2}$$

PDF $f_Y(y)$

$$f_Y(y) = \begin{cases} \frac{1}{4}, & -2 < y < -1 \\ \frac{1}{4}, & 1 < y < 2 \\ 0, & \text{otherwise} \end{cases}$$

with point mass $P(Y = 0) = \frac{1}{2}$

Q7. Binomial Approximation of Stock Movement Condition

$$S_{1000} = S_0 u^k d^{1000-k}$$
$$u^k d^{1000-k} \geq 1.30$$

Log Transformation

$$k(\ln u - \ln d) + 1000 \ln d \geq \ln(1.30)$$

Solve for k

$$k \geq \frac{\ln(1.30) - 1000 \ln d}{\ln u - \ln d} \approx 469.4$$
$$k \geq 470$$

Distribution

$$k \sim \text{Binomial}(1000, 0.52)$$

Final Probability

$$P(k \geq 470) = \sum_{k=470}^{1000} \binom{1000}{k} (0.52)^k (0.48)^{1000-k}$$

Q8. Exponential Distribution

Given

$$\lambda = \frac{1}{2}, \quad f(x) = \frac{1}{2}e^{-x/2}$$

(a)

$$P(X > 2) = e^{-1} \approx 0.3679$$

(b)

$$P(X \geq 10 | X > 9) = \frac{e^{-5}}{e^{-9/2}} = e^{-1/2}$$

Q9. Distribution of $Z = \sqrt{X^2 + Y^2}$

CDF

$$\begin{aligned} F_Z(z) &= P(X^2 + Y^2 \leq z^2) \\ &= \int_{-z}^z \int_{-\sqrt{z^2-x^2}}^{\sqrt{z^2-x^2}} f_X(x) f_Y(y) dy dx \end{aligned}$$

PDF

$$f_Z(z) = \frac{d}{dz} F_Z(z)$$

Using Leibniz rule:

$$\frac{d}{dz} \sqrt{z^2 - x^2} = \frac{z}{\sqrt{z^2 - x^2}}$$

Final Result

$$f_Z(z) = z \int_{-z}^z f_X(x) \frac{f_Y(\sqrt{z^2-x^2}) + f_Y(-\sqrt{z^2-x^2})}{\sqrt{z^2-x^2}} dx, \quad z > 0$$

Q10. Joint Distribution of M_n and M_{n+1}

Definitions

$$M_n = \max(X_1, \dots, X_n)$$

$$P(M_n \leq x) = F(x)^n$$

Joint CDF

Case 1: $x \leq y$

$$P(M_n \leq x, M_{n+1} \leq y) = F(x)^n F(y)$$

Case 2: $x > y$

$$P(M_n \leq x, M_{n+1} \leq y) = F(y)^{n+1}$$

Final Expression

$$P(M_n \leq x, M_{n+1} \leq y) = \begin{cases} F(x)^n F(y), & x \leq y \\ F(y)^{n+1}, & x > y \end{cases}$$

End of Lecture Scribe

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